

HERNANDO, PANTINE B.

BSIT-3

CHALLENGE - 8 - CRYPTOGRAPHY

substitution1 



Medium Cryptography picoCTF 2022 Substitution_cipher

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Hints 

Description

1 2

A second message has come in the mail, and it seems almost identical to the first one. Maybe the same thing will work again.

Download the message [here](#).

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picoCTF{FLAG}

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★ SEARCH A TOOL ON dCODE

e.g. type 'boolean'

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Mono-alphabetic Substitution

Tool to decrypt monoalphabetical substitution and find each letter of a substituted message with a deranged alphabet (monoalphabetic cipher).

Mono-alphabetic Substitution - [dCode](#)

Tag(s) : Substitution Cipher

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dCode is free and its tools are a valuable help in games, maths, geocaching, puzzles and problems to solve every day!
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MONO-ALPHABETIC SUBSTITUTION

Cryptography > Substitution Cipher > Mono-alphabetic Substitution

MONOALPHABETIC SUBSTITUTION DECODER

★ ALPHABETIC SUBSTITUTION CIPHERTEXT

LKOb (bwvek ove 1gqkhej kwj osgx) gej g kyaj vo lvrqhkje
bjlhetky lvrqjktktvu. Lvukjbkgukb gej qejbjukjz dtkw g
bjk vo lwgssjuxjb dwtlw kjbk kwjte lejgktftky, kjlwutlgs
(guz xvxtstux) bitssb, guz qevmsjr-bvsftux gmtstky.
Lwgssjuxjb hhhgssy lyfje g uhrmje vo lgkxvetjb, guz
dwtu bvstiz. ialw vtiszb a bketux (lassiz a osgx) dwtlw

★ SPACES ARE RELEVANT AND MUST BE KEPT (ARISTOCRAT CIPHER)
☐ CAN BE IGNORED OR ARE MISSING (PATRISTOCRAT CIPHER)

★ PLAINTEXT LANGUAGE French (Français)

► DECRYPT AUTOMATICALLY

OTHER DECRYPTION METHODS

☒ KNOWING THE SUBSTITUTION ALPHABET

QWERTYUIOPASDFGHJKLZXCVBNM

☐ KNOWING THE CODED/RECIPROCAL ALPHABET

☐ KNOWING THE KEYWORD USED TO GENERATE THE ALPHABET

☐ ATTACK WITH PROBABLE WORD/KNOWN PLAIN TEXT SUBSTITUTION

☐ MANUAL DECRYPTION (EMPTY VISUAL TOOL)

► DECRYPT

See also: [Word Desubstitution/Pattern](#) — [Frequency Analysis](#)

MONOALPHABETIC SUBSTITUTION ENCODER

★ UNSUBSTITUTED PLAIN TEXT

dCode is able to decipher an alphabetic substitution of a text by analyzing the frequencies of occurrence of the letters. The message must be as long as possible so that dCode can accurately find the substitution alphabet.

★ SUBSTITUTION ALPHABET QWERTYUIOPASDFGHJKLZXCVBNM

► ENCRYPT

Summary

- ★ Monoalphabetic Substitution Decoder
- ★ Monoalphabetic Substitution Encoder
- ★ Custom Deranged Alphabet Generator
- ★ What is a (mono-)alphabetical substitution? (Definition)
- ★ How to encrypt using an alphabetical substitution?
- ★ What do K1, K2, K3, K4 mean?
- ★ How to decrypt using an alphabetical substitution?
- ★ How to recognize a mono alphabetical substituted text?
- ★ How to decipher a substitution without the alphabet?
- ★ What is the MCMC technique?
- ★ What are the variants of the substitution cipher?

Similar pages

- ★ Word Desubstitution/Pattern
- ★ Deranged Alphabet Generator
- ★ Caesar Cipher
- ★ Word Substitution

This challenge is a medium-difficulty cryptography task from **picoCTF 2022** that focuses on breaking a monoalphabetic substitution cipher without an explicit key.

Description

The challenge provides a text file containing a message where every letter of the alphabet has been replaced by another. Unlike the previous challenge in the series, the substitution key is not provided at the top of the file.

Ciphertext Snippet:

LKOb (bwvek ove 1gqkhej kwj osgx) gej g kyaj vo Turghkje bj Thetky lvrqjktktvu. Lvuk...

Solution Strategy

The core method for breaking a substitution cipher of this length is **Frequency Analysis**. This technique relies on the fact that in any given language, certain letters and letter combinations appear with predictable frequencies.

Steps taken to solve:

1. **Analyze the Text:** By looking at the ciphertext, we can identify common patterns. For example, the first word LKOb followed by a bracketed phrase suggests a standard introduction format.

Search for a tool

SEARCH A TOOL ON dCODE

e.g. type 'boolean'

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Results

dCode tried to find the correct alphabet and its substitution automatically. The result is a draft that should allow you to perform the decryption manually by indicating letters in each cell.

CTFS (SHORT FOR CAPTURE THE FLAG) ARE A TYPE OF COMPUTER SECURITY COMPETITION. CONTESTANTS ARE PRESENTED WITH A SET OF CHALLENGES WHICH TEST THEIR CREATIVITY, TECHNICAL SKILLS, AND PROBLEM-SOLVING ABILITY. CHALLENGES USUALLY COVER A NUMBER OF CATEGORIES, AND WHEN SOLVED, EACH YIELDS A STRING (CALLED A FLAG) WHICH IS SUBMITTED TO AN ONLINE SCORING SERVICE. CTFS ARE A GREAT WAY TO LEARN A WIDE ARRAY OF COMPUTER SECURITY SKILLS IN A SAFE, LEGAL ENVIRONMENT, AND ARE HOSTED AND PLAYED BY MANY SECURITY GROUPS AROUND THE WORLD FOR FUN AND PRACTICE. FOR THIS PROBLEM, THE FLAG IS: PICOCTF{FR3XU3NCY_4774CK5_4R3_C001_6E0659FB}

1 GMLZJQXWNTISRUVCBKHFDAYP

2 XSQWRVAUKETCBJFZPMLINOHGYD

Mono-alphabetic Substitution - dCode

Tag(s) : Substitution Cipher

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2. **Frequency Mapping:** In English, the most common letters are E, T, A, O, I, N. By counting the most frequent characters in the ciphertext (like 'j', 'v', or 'g'), we can begin mapping them to these common English letters.

3. **Pattern Recognition:** We look for short, common words. A three-letter word appearing frequently might be "the." In this ciphertext, the word kwj appears in positions where "the" would logically fit.

4. **Automated Decryption:** Using a tool like dCode's Mono-alphabetic Substitution Decoder, the ciphertext was processed to find the most likely substitution alphabet. The tool analyzes letter frequencies and bigrams (common two-letter pairs) to reconstruct the original text.

SECURITY SKILLS IN A SAFE, LEGAL ENVIRONMENT, AND ARE HOSTED AND PLAYED BY MANY SECURITY GROUPS AROUND THE WORLD FOR FUN AND PRACTICE. FOR THIS PROBLEM, THE FLAG IS: PICOCTF{FR3XU3NCY_4774CK5_4R3_C001_6E0659FB}

1 GMLZJQXWNTISRUVCBKHFDAYP

2 XSQWRVAUKETCBJFZPMLINOHGYD

Mono-alphabetic Substitution - dCode

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Y, TECHNICAL (AND GOOGLING)

BITSSB, GUZ QEVMSJR-BVSF

SKILLS, AND PROBLEM-SOLV

TUX GMTSTKY, LWGSSJUXJB HB

ING ABILITY, CHALLENGES US

HGSSY LVFJE G UHRMJE VO LG

UALLY COVER A NUMBER OF CA

KJXVETJB, GUZ DWJU BVSFJZ,

TEGORIES, AND WHEN SOLVED,

JGLW YFJSZB G BKETUX (LGS

EACH YIELDS A STRING (CAL

SJZ G OSGX) DWTLW TB BHMRT

LED A FLAG) WHICH IS SUBMI

KKJZ KV GU VUSTUJ BLVETUX

TIED TO AN ONLINE SCORING

BJEFTLJ, LKOB GEJ G XEJGK

SERVICE, CTFS ARE A GREAT

DGY KV SJGEU G DTZJ GEEGY

WAY TO LEARN A WIDE ARRAY

VO LVROHKJE BJLHETKY BITSS

OF COMPUTER SECURITY SKILL

B TU G BGOJ, SJXGS JUFTEVU

S IN A SAFE, LEGAL ENVIRON

RJUK, GUZ GEJ NVBKJZ GUZ Q

MENT, AND ARE HOSTED AND P

SGYJZ MY RGUY BJLHETKY XEV

LAYED BY MANY SECURITY GRO

HQB GEVHUZ KWJ DVESZ OVE O

UPS AROUND THE WORLD FOR F

HU GUZ QEGLKTLJ, OVE KWTB

UN AND PRACTICE, FOR THIS

QEVMSJR, KWJ OSGX TB: QTLV

PROBLEM, THE FLAG IS: PICO

LKO{OE3AH3ULY-4774LIS-4E3-

CTF{FR3XU3NCY-4774CK5-4R3-

L001-6J0659OM}

C001-6E0659FB}

alphabet?

What is the MCMC technique?

What are the variants of the substitution cipher?

Similar pages

Word Desubstitution/Pattern

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Word Substitution

Atbash Cipher

Frequency Analysis

Substitution Cipher

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Keywords

substitution, monoalphabetical, alphabet, cryptogram, aristocrat, patristocrat, pangram, correspondence, mapping

Links

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Final Result

After fully decrypting the message, the text concludes by explicitly stating the flag.

Decrypted Flag:

PICOCTF{FR3QU3NCY_4774CK5_4R3_C001_6E0659FB}